

CS471 – Lesson 20 In-Class Worksheet

Probabilistic Systems: Bayesian Networks

Part 1: Guided In-Class Worksheet (Instructor-Led)

Objective: Represent systems as Directed Acyclic Graphs (DAGs), trace family relationships, evaluate conditional independence, and factor joint probability distributions.

1. Bayesian Network Structure & Family Relationships

Consider the tactical air-strike planning network from Quiz V:

$$W \rightarrow V, \quad W \rightarrow C, \quad V \rightarrow M, \quad C \rightarrow M$$

where W is Weather, V is Optical Visibility, C is SATCOM Communications, and M is Mission Execution.

a. Sketch or diagram the Directed Acyclic Graph (DAG) for this network.

b. List the immediate **parents** of M : _____

c. Is W an **ancestor** of M ? Circle one: **YES** / **NO**
Briefly justify your answer:

d. Complete the conditional-independence statement: $V \perp C \mid$ _____
Explain why observing this variable renders V and C conditionally independent.

2. Joint Distribution Factorization

- Write the mathematical chain rule factorization for the joint distribution $P(W, V, C, M)$ using the parents of each node.
- Calculate the total number of independent probability parameters required to populate this Bayesian network versus an unconstrained full joint probability table.

Part 2: Student Independent Practice

Objective: Analyze network topology, determine conditional independence, and compute tactical joint probabilities.

A. Autonomous Air Defense Network

An integrated air defense system (IADS) features five automated nodes:

- R : Adversary Radar Jamming
- T : Low-Altitude Terrain Clutter
- D : Drone Detection Radar
- I : Infrared Optical Sighting
- F : Surface-to-Air Missile (SAM) Fire Authorization

The network topology is defined by the edges:

$$R \rightarrow D, \quad T \rightarrow D, \quad T \rightarrow I, \quad D \rightarrow F, \quad I \rightarrow F$$

1. List all immediate **parents** of D .
2. List all **descendants** of T .
3. Complete the conditional independence statement: $D \perp I \mid \underline{\hspace{2cm}}$.
4. Write the full joint probability decomposition $P(R, T, D, I, F)$ in terms of local conditional distributions.

B. Joint Probability Computation

Using the air defense network from Problem A, suppose reconnaissance yields the following parameters:

- $P(R = 1) = 0.30$
- $P(T = 1) = 0.40$
- $P(D = 1 \mid R = 1, T = 1) = 0.10$
- $P(I = 1 \mid T = 1) = 0.70$
- $P(F = 1 \mid D = 1, I = 1) = 0.90$

Compute the joint probability that all five events occur simultaneously:

$$P(R = 1, T = 1, D = 1, I = 1, F = 1)$$